ADVANCING ABDOMINAL AORTIC ANEURYSM RISK ASSESSMENT THROUGH ANEURYSM MORPHOLOGICAL CHARACTERIZATION

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Introduction

Abdominal aortic aneurysm (AAA) is the localized dilation of the infrarenal aorta, whose rupture causes thousands of deaths worldwide each year, with a higher rupture rate reported in women [1]. Therefore, beyond the usually adopted maximum diameter criterion, more reliable methods to estimate AAA rupture risk are needed [1].

Towards this ultimate goal, this work develops a computational pipeline which, starting from the automatic isolation of the aneurysmal sac (AS) in patient-specific 3D vascular models, extracts a set of relevant morphological features to advance the geometrical characterization of AAAs [2, 3]. In this regard, particular emphasis is placed on geometrical parameters capturing the three-dimensional nature of vascular morphology [4].

Methods

This study is conducted on a clinical cohort comprising 39 patients, including 22 with unruptured AAAs and 17 with ruptured AAAs.

For each patient, the 3D surface of the infrarenal abdominal aorta is reconstructed through a semi-automated segmentation process and subsequently provided as input to the software tool under development.

The tool first computes the centerline of the aortic geometry. The vessel is then sectioned at regular intervals by planes orthogonal to the centerline, allowing the diameter to be determined at each cross-section. For each cross-section position, the corresponding radius length is recorded in a two-dimensional radius-position plane, generating a scatter plot. The interpolation of data points yields a function, $R(x) \in C^2$, expressing the radius variation with respect to the continuous position x along the centreline (Figure 1a). The AS isolation procedure relies on this one-dimensional description of the aneurysm through the R(x) function. The AS is defined as:

$$AS = \{x : x_{Left} \leq x < x_{Rigth}\},$$

where, x_{Left} is the location of the local minimum of R(x) preceding the location of the local maximum of R'(x) upstream of the peak of R(x) and x_{Rigth} is the location of the local minimum of R(x) following the location of the local negative minimum of R'(x) downstream the peak.

Once the AS is isolated, the computational pipeline extracts a set of geometrical descriptors previously identified in the literature as accurate predictors of AAA rupture risk together with less investigated morphological parameters [3, 4]. In particular, parameters such as centerline curvature and torsion [4] provide further insight into the three-dimensional morphology of the aneurysm.

Results

Figure 1 shows the outputs of the computational pipeline for a representative case, illustrating the AS isolation procedure and the subsequent extraction of centerline curvature.

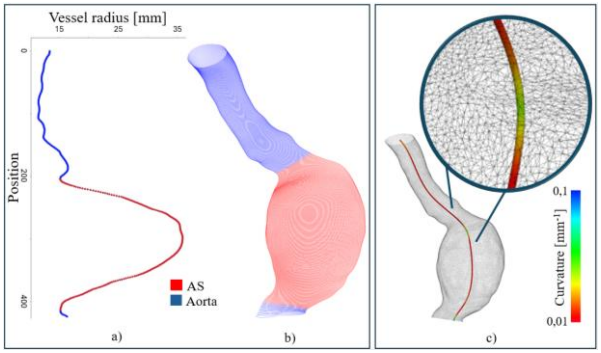


Figure 1: a) Radius-position plot for a representative case, with data points and the interpolated curve R(x); b) Sliced surface of the illustrative case; c) Centerline curvature distribution.

Conclusions

The results show that the identification strategy implemented in the software tool successfully detects the AS from the proximal to the distal neck [2]. These early findings suggest that the proposed approach achieves a level of AS detection accuracy comparable to that of three-dimensional methods, while retaining the computational efficiency inherent to one-dimensional techniques.

Future work will focus on the development of an AAA rupture risk predictor based on the statistical integration of the geometrical descriptors extracted by the software, biomechanical indicators derived from structural mechanics and fluid dynamics analyses, and relevant clinical variables [5].

References

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